

The Hyperframe Program: Accumulating Interpretations

A running interpretive companion to the mechanized library

Project `thejeb`

July 29, 2026

Abstract

This is a working log, not a finished paper. As each result is mechanized in Lean and merged to the main library, we record here *what it means* — the interpretation the machine-checked statement licenses — so that the interpretations accumulate rather than evaporate. The discipline is fixed: every theorem is cited by its exact Lean identifier and is proved (sorry-free, no `native.decide`; axioms `propext`, `Classical.choice`, `Quot.sound` only); the surrounding prose is interpretation, ours to defend, and is marked as such. The frame of reference throughout is the hyperframe stance developed in the companion epistemology note: set theoretic estimation [2] studies *coherence, not truth*, over set-valued (possibility-set) frames extracted from context-bounded texts. The information-lattice thread connects to Shannon’s lattice of information as synthesized by Delsol et al. [3]; the set-of-sets thread extends the STE-for-decryption line of Carlson [1].

1 The backbone (established earlier)

Three prior machine-checked facts anchor everything below and are used as load-bearing lemmas.

- **2-locality of frames.** `frame_hasHelly_two (Ste.HellyConsistency)`: for frame corpora, pairwise consistency forces global consistency. Coherence is a purely 2-local property — the metric-free discriminator of feasible structure.
- **The coupling dichotomy.** `rank_submodular` and `couplingExcess_pos_of_not_le (Ste.PartitionRank/S` the partition rank is submodular, and the coupling excess is zero exactly when readings decompose and positive exactly when they genuinely entangle.
- **The lattice is metric.** `shannonDist_triangle (Ste.InfoDistance)`: the reading lattice is a metric space under the counting Shannon distance, the triangle inequality falling out of submodularity.

Each new thread below is read against this backbone.

2 The set-valued frame: the primitive, made precise

Module `Ste.SetValuedFrame`. The primitive of the whole program is now a first-class object. A *set-valued frame* assigns each variable a possibility set; its induced constraint is the box `propertySet F = Set.univ.pi F (propertySet_isRectangle)`, hence it glues with no obstruction over every cover (`propertySet_cechVanishesCover`). A corpus is feasible exactly when every variable’s possibility-set intersection is nonempty (`corpusFeasibilitySet_nonempty_iff`).

Interpretation. This is the formal content of “reference is underdetermined but bounded.” A source does not name a referent; it fences a set of admissible ones, and the arithmetic of meaning is then *coordinatewise set intersection*. The crisp frame is recovered as the singleton special case (`ofCrisp`), with 2-locality transported (`hasHelly_ofCrisp_two`) — so the classical determinate-reference model is not discarded but embedded as the zero-uncertainty corner of the set-valued one. That the corpus feasibility set is a formal-concept extent (`isExtent_corpusFeasibilitySet`) says the possibility-set picture and the concept-lattice picture are the same object seen twice.

3 Local, non-tight gluing: coherence has a shape, not a verdict

Module `Ste.LocalGluing`. Gluing is no longer all-or-nothing. `GluesOn T U J'` says a compatible family assembles over a *sub-cover* J' ; the `gluingComplex` is the family of sub-covers on which it does, and it is downward closed (`mem_gluingComplex_of_subset`) — an abstract simplicial complex. The tight sheaf condition is exactly the top face being present (`cechVanishesCover_iff_forall_univ_mem_gluingComplex` `cechVanishesCover_iff_gluesLocallyAt_card`). The diagonal is the separating witness: it glues locally at width one but not at two (`diagonal_gluesLocallyAt_one`, `diagonal_not_gluesLocallyAt_two`, `diagonal_local_not_tight`).

Interpretation. Incoherence is not a single bit. A corpus that fails to cohere globally still coheres over a rich, structured family of sub-corpora, and that family — the gluing complex — is the real object of study. This is the paraconsistent stance made geometric: when the whole does not glue, the meaningful content is the complex of pieces that do, and the obstruction is located, not merely detected. The bridge `cechVanishesCover_of_hasHelly_gluingFiber` ties this back to the backbone: enough local Helly structure on the fibers forces the top face, i.e. local coherence climbs to global. “Local but not tight” was the right instinct; it names the strictly weaker middle between per-piece consistency and global solvability.

4 Multi-document coreference: identity as a lattice limit

Module `Ste.Coreference`. Coreference across documents is now modeled. Mentions live over disjoint documents; each document contributes a local coreference; the corpus-forced coreference is the setoid-lattice join of those contributions with the closure of the cross-document links (`forcedCoref`), characterized as the least adequate partition (`forcedCoref_eq_sInf`). It is monotone in evidence (`forcedCoref_mono_docs`, `forcedCoref_mono_link`, `rank_forcedCoref_mono_docs`), it never crosses a document boundary without a link (`forcedCoref_no_links_same_doc`), and a two-link chain can force a cross-document identification (`witness_cross_doc_forced`).

Interpretation. “Who is the same as whom, across a corpus” is not a search problem with an answer to be guessed; it is a *fixed point* of constraint accumulation — the least partition adequate to all the local readings and links. Identity is derived, not posited, and it can only sharpen as documents arrive (monotonicity). The locality lemma is the honest guardrail: the formalism refuses to invent cross-document identity out of thin air — it merges two mentions only when a chain of evidence links them. This is coreference as the partition-lattice meet, the same `sInf` structure the backbone’s forced-reading result already used, now indexed by document provenance.

5 Fuzzy frames: possibility with a gradient

Module `Ste.FuzzyFrame`. The possibility set acquires a degree. A `FuzzyFrame` is a membership $\Xi \rightarrow [0, 1]$; intersection is the min t-norm (`FuzzyFrame.inter`); the α -cut recovers crisp sets

(`FuzzyFrame.cut`). Feasibility becomes graded — a `consistencyDegree` — and the two levels are tied at every threshold by `cut_fuzzyFeasibility` (the α -cut of fuzzy feasibility equals the crisp feasibility of the α -cuts). The crisp theory is the boundary: `consistencyDegree_crisp_eq_one_iff` (a crisp corpus scores 1 iff feasible), and the graded Helly-2 analogue `weightedFrame_gradedHelly_two` holds.

Interpretation. This answers “is a hyperframe really a fuzzy powerset with a density?” in the possibilistic direction (Zadeh, Dubois–Prade): yes, and the crisp set-valued frame is exactly its $\{0, 1\}$ -valued restriction. The α -cut theorem is the load-bearing one — it means the graded theory is not a separate universe but a *stack* of crisp theories indexed by confidence threshold, so every crisp result (2-locality, gluing, feasibility) has an immediate graded shadow obtained by cutting. The genuine *probabilistic* direction — a normalized density with Shannon entropy on top — remains out of reach in Lean (no general discrete Shannon entropy in Mathlib); the honest boundary from the epistemology discussion stands.

6 Carlson set-of-sets: lifting a set breaks 2-locality

Module `Ste.Carlson.SetOfSets`. A second-order extension of Carlson-style STE: a source offers not a possibility set but a *set of* possibility sets, “the answer is one of these sets.” Consistency becomes selectability — a system of representatives with nonempty intersection (`SelectableConsistent`). The singleton family recovers the first-order theory exactly (`selectable_singleton_iff`, `hasHellySel_singleton`). The headline: a `Fin 3` triangle of two-element families is pairwise selectable yet has no global selection (`exists_not_hasHellySel_two`), so the second-order Helly number is at least three.

Interpretation. This is the sharpest “the Helly number would change” result. Uncertainty *about which possibility set applies* is a genuinely higher-order uncertainty than uncertainty *within* a possibility set, and the jump is visible in exactly one invariant: the first order is 2-local (`frame_hasHelly_two`), the second order is not. Operationally this is a correctness cliff: any solver that ingests set-of-sets data and relies on the linear pairwise scan is unsound; a $k \geq 3$ check is mandatory. Interpretively it says the “maybe we do not know which reading” move — disjunction over readings — is not free; it costs locality, which is the very property that made the first-order theory tractable.

7 Information-lattice geometry: coherence has a curvature

Module `Ste.InfoTopology`. The honest first move is to kill the trivial reading: on a finite carrier the Shannon metric is uniformly discrete (`one_le_shannonDist_of_ne`), so the point-set *topology* is nothing. The content is *geometric*, and it is a dichotomy between the two lattice operations. **Join is a contraction** and jointly 1-Lipschitz (`shannonDist_sup_right_le`, `shannonDist_sup_le_add`): combining information is metrically stable. **Meet is not** (`shannonDist_inf_not_contraction`, a `Fin 4` witness): a tiny change in two readings can produce an arbitrarily large change in their common information. In fact meet has *no* Lipschitz constant at all: for every C the $2k$ -carrier witness ($k = C + 1$; top vs. halves against the perfect matching) gives $d(P, Q) = 1$ yet $d(P \sqcap R, Q \sqcap R) = k$ (`meet_not_lipschitz`, `Ste.MeetScaling`) — the general- k theorem, no longer just the $k = 2$ instance. And the metric, though bipartite-parity (`shannonDist_add_rank_add_rank`), embeds in no ℓ_1 cube (`pentagonal_violation`, already at $|\alpha| = 3$) and is not a median space (`median_not_unique`); it is the covering-graph distance of a geometric lattice — geodesic along refinement chains (`aligned_of_chain`, `le_sup_of_aligned`), submodular-rank controlled, but non-cubical.

Interpretation. The join/meet asymmetry is the geometric face of the whole program’s recurring theme. *Aggregating* evidence (join, ‘and all these readings together’) is well-conditioned — small perturbations of the inputs move the consensus only a little, which is exactly what a trustworthy accumulation procedure needs, and it is why corpus growth slides the STE consensus along a geodesic rather than lurching. *Extracting what readings share* (meet, common information) is ill-conditioned: this is the counting-shadow of the Gács–Körner discontinuity of common information, and it is a warning — “what do these sources agree on, exactly” is a numerically unstable question, and any procedure that leans on it inherits that instability. The non- ℓ_1 , non-median results say the geometry is genuinely its own: coherence-space is not a Hamming cube one can coordinate away, even in the crisp counting shadow. The single open direction is asymptotic — normalized, these finite discrete spaces should converge to the measurable-partition space under the Rokhlin metric (Delsol’s D in ergodic-theory clothing), where the topology is finally non-trivial; that is not a near-term Lean target and is flagged as such.

Toward a paper

All six faces are now machine-checked, and the throughline is a single claim: *the hyperframe — a bounded but underdetermined, possibly graded, possibly higher-order set of admissible readings — carries exactly the structure needed to reason about coherence without truth, and that structure is a lattice with a metric, a gluing complex, and a sharp locality invariant that degrades and lifts in controlled ways.* Set-valued frames give the primitive; local gluing gives coherence a shape (the gluing complex); coreference gives identity as a lattice limit; fuzzy frames add the gradient and degrade to crisp at each threshold; the set-of-sets result marks the locality cliff (Helly $2 \rightarrow \geq 3$); and the metric geometry gives it a curvature — aggregation (join) well-conditioned, common-information (meet) not. Each face is a theorem; the paper is their assembly, and its spine is the join/meet conditioning asymmetry, which recurs as the tractable/hard, decomposed/ coupled, and gluable/obstructed dichotomy under three other names.

Proposed paper skeleton. (i) The hyperframe primitive and coherence-not-truth (epistemology note as §1–2); (ii) the feasibility lattice and 2-locality (backbone + set-valued frame); (iii) coherence as gluing, local to tight (gluing complex, Čech obstruction); (iv) the geometry of coherence (Shannon metric, join/meet conditioning); (v) the controlled deformations — graded (fuzzy) and higher-order (set-of-sets, the locality cliff); (vi) identity across documents (coreference); (vii) the honest boundary (fidelity vs consistency, the entropy ceiling). Everything cited by Lean name; nothing claimed beyond the kernel.

References

- [1] Albert H. Carlson. *Set Theoretic Estimation Applied to the Information Content of Ciphers and Decryption*. Ph.d. dissertation, University of Idaho, Moscow, Idaho, USA, May 2012. Retrieved via Internet Archive snapshot (2024-07-13) of the author’s ResearchGate full text, publication 349064680.
- [2] Patrick L. Combettes. The foundations of set theoretic estimation. *Proceedings of the IEEE*, 81(2):182–208, 1993. doi: 10.1109/5.214546. Retrieved from the author’s page, <https://pcombet.math.ncsu.edu/proc.pdf>.
- [3] Idris Delsol, Olivier Rioul, Julien Béguinot, Victor Rabet, and Antoine Souloumiac. An information theoretic condition for perfect reconstruction. *Entropy*, 26(1):86, 2024. doi:

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